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Grid Bottlenecks: The Underappreciated Capex Cycle in Electrical Equipment

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ABSTRACT

Transmission and distribution infrastructure, rather than generation capacity itself, is emerging as the binding constraint on electrification and data-center buildout. This paper documents the resulting multi-year capital expenditure cycle for grid equipment manufacturers and assesses which segments of the value chain are best positioned to capture it.

KEY FINDINGS

01 Transformer lead times have extended materially versus pre-2020 norms, with capacity additions running behind demand from electrification and data-center buildout.

02 Interconnection queues for new generation capacity are now dominated by wait times for grid studies and required upgrades rather than by construction timelines themselves.

03 Electrical equipment manufacturers with vertically integrated component supply appear better positioned to capture pricing power as the cycle progresses.

04 Utility capital expenditure guidance has been revised higher across several consecutive planning cycles, suggesting durable rather than transitory demand.

05 Earnings visibility in grid equipment now extends further out than is typical for industrial capital goods, supported by order backlogs rather than forecasts.

DISCUSSION

The paper draws on utility capital plans, interconnection queue data, and equipment manufacturer order books to characterize the shape and likely duration of the grid capex cycle.

It concludes with a framework for differentiating manufacturers by exposure to the most capacity-constrained equipment categories, and by their ability to pass through input cost inflation given backlog composition.